

HIGH-LEVEL DESIGN

Agentic Loan Origination System

Loan Against Property (LAP)

Conversational onboarding, document intelligence, fraud and credit risk models, and a deterministic FOIR / LTV eligibility engine, running on an on-premises Ollama LLM.

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1. Introduction

1.1 Purpose

This document describes the high-level design (HLD) of an agentic Loan Origination System (LOS) for **Loan Against Property (LAP)**. It defines the overall architecture, the major components and their responsibilities, how data and decisions flow between them, and the key technology, security and compliance choices. It is intended to align stakeholders before detailed design and build begin.

1.2 Scope

In scope

- A conversational assistant (chatbot) for applicant onboarding, document collection and status queries, powered by an on-premises Ollama LLM.
- Document intelligence: classification, OCR, structured extraction and validation of KYC, income and property documents.
- Fraud and anomaly detection across identity, income, documents and property.
- Credit risk models: Probability of Default (PD), Loss Given Default (LGD), Exposure at Default (EAD) and Expected Loss (EL).
- Property assessment: legal, technical and valuation inputs.
- A deterministic policy engine that computes FOIR-based and LTV-based eligibility and the final eligible loan amount.
- Decisioning (auto-approve / refer / reject) with reason codes, human review and a full audit trail.
- Early Warning System (EWS) design for post-disbursement monitoring (later phase).

Out of scope for this document

- Loan servicing, collections and general ledger integration.
- Detailed API contracts, database schemas and UI designs (covered in low-level design).
- Model development detail such as exact features and hyperparameters.

1.3 Audience

Solution and enterprise architects, engineering leads, data science and credit risk teams, compliance and information security reviewers, and product owners.

1.4 Assumptions

- Ollama is already installed and will be the LLM runtime. The specific model and GPU hardware are yet to be confirmed.
- All applicant data, including Aadhaar, PAN and bank statements, stays within the organisation's own infrastructure.
- Credit policy parameters (FOIR, LTV caps, tenure, rates, ticket size) are owned by the Credit team and will be supplied as configuration, not hard-coded.
- Historical loan data with repayment performance may or may not be available for model training. The design supports both cases (see Section 11).
- Items marked **[TBC]** are open and need confirmation. They are collected in Section 18.

2. Business Context and Objectives

LAP is secured lending against residential, commercial or industrial property, often used for business needs. Two risk streams determine the outcome and are treated as separate concerns in this design:

- **Borrower risk:** the ability and willingness to repay (income, obligations, bureau behaviour, bank conduct).
- **Property risk:** the ability to recover dues if the borrower defaults (title, valuation, marketability, legal complexity, time and cost to foreclose).

2.1 Objectives

Objective	Description	Indicative measure [TBC]
Faster turnaround	Reduce the time from application to sanction by automating intake, extraction and checks.	Median time to conditional sanction
Consistent, auditable decisions	Every decision reproducible from logged inputs, model versions and policy version.	100% of decisions with reason codes and full input snapshot
Lower fraud and credit loss	Catch document forgery, income inflation and duplicate mortgages early.	Fraud catch rate, early-default rate
Better applicant experience	One conversational entry point for onboarding, uploads and status.	Completion rate, drop-off by stage
Data privacy by design	Sensitive data processed on-premises with strict access control.	Zero PII sent to external LLM services

3. Architecture Principles

Principle	Implication for the design
LLM assists, never decides	The LLM handles conversation, extraction and explanation. Eligibility, approval and rejection come only from the deterministic policy engine, decision service and scored models. The LLM cannot change application status or amounts.
Deterministic core, probabilistic edges	FOIR, LTV and all amount calculations are code with unit tests. ML models produce scores that are inputs to rules, not final answers.
Treat documents as untrusted data	Text extracted from uploads is data only. It is never allowed to act as instructions to the LLM (prompt-injection defence).
Separation of borrower and property risk	PD and LGD are modelled and reported separately, then combined into EL. Components stay visible to underwriters.
Configuration over code for policy	Rules, thresholds, LTV caps and FOIR limits live in versioned configuration owned by Credit, with maker-checker change control.
Explainable and auditable by default	Each decision stores inputs, model and policy versions, scores, reason codes and the human actions taken.
Fail safe to human review	Low-confidence extraction, missing data, service failures and conflicting signals route to a human, never to auto-approval.

Principle	Implication for the design
Data stays on-premises	Model inference, OCR and storage run inside the organisation's network. External calls are limited to regulated data providers.

4. Logical Architecture

The system is organised into six layers. Requests flow from channels down through the agent and orchestration layer into deterministic intelligence and decision services, which use external integrations and a shared data platform.

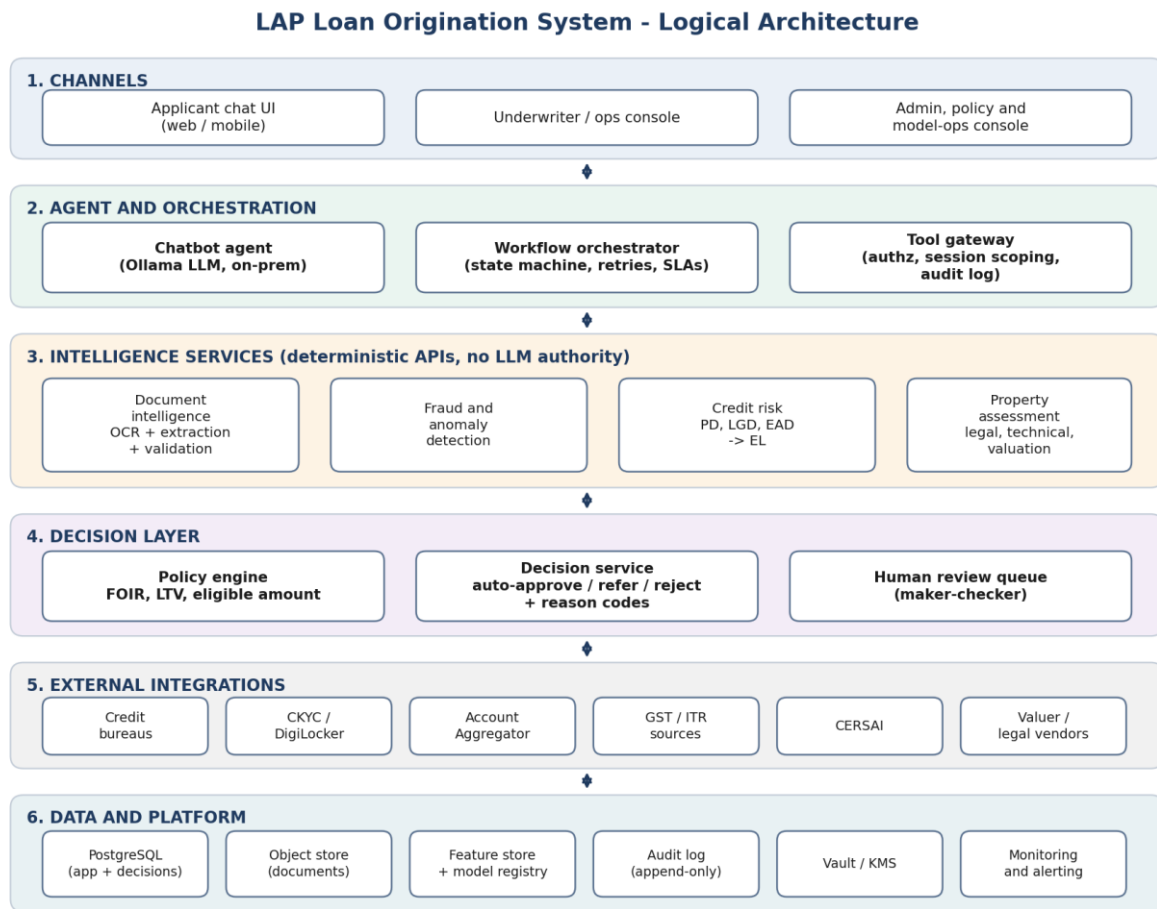


Figure 1: Logical architecture of the LAP LOS

Layer	Responsibility
1. Channels	Applicant chat UI, underwriter / operations console, and admin console for policy, model and configuration management.
2. Agent and orchestration	Ollama-based chatbot agent, workflow orchestrator that owns application state, and a tool gateway that is the only path from the LLM to system actions.
3. Intelligence services	Document intelligence, fraud and anomaly detection, credit risk models (PD, LGD, EAD to EL), and property assessment. All are deterministic APIs.
4. Decision layer	Policy engine (FOIR, LTV, eligible amount), decision service with reason codes, and a human review queue with maker-checker.

Layer	Responsibility
5. External integrations	Credit bureaus, CKYC and DigiLocker, Account Aggregator, GST and ITR sources, CERSAI, and valuer / legal vendors.
6. Data and platform	Relational store, object store for documents, feature store and model registry, append-only audit log, secrets management, monitoring.

5. Component Design

5.1 Chatbot Agent (Ollama)

The chatbot is the applicant's single entry point. It collects information conversationally, guides document upload, and answers status questions. Ollama serves the LLM through its local API, including tool calling and JSON-schema constrained output.

Responsibilities

- Slot-filling: applicant type (salaried / self-employed / business), property details, loan purpose, requested amount and tenure, co-applicants and property owners.
- Explaining which documents are needed and why, and confirming receipt.
- Answering status queries and narrating decisions using data returned by tools.
- Handing off to a human agent on request, on repeated confusion, or on sensitive topics.

Tools exposed to the agent (all served through the tool gateway, all scoped to the current applicant session):

Tool	Purpose	Notes
create_application	Start a new LAP application	Returns application ID
update_application_field	Save a validated slot value	Schema-validated, logged
get_required_documents	List documents required for this applicant and property type	Driven by product configuration
request_document_upload	Generate a secure upload link or slot	No file content passes through the LLM prompt
get_application_status	Return current stage and pending items	Read-only
get_eligibility	Return indicative eligibility from the policy engine	LLM must quote values verbatim, never compute
explain_decision	Return reason codes and plain-language text	Only approved reason-code wording
handoff_to_human	Create a callback or live-agent ticket	Includes conversation summary

Rule: The agent never calculates, estimates or promises a loan amount. Any figure shown to an applicant comes from the get_eligibility tool and is labelled indicative until formal sanction.

5.2 Workflow Orchestrator

The orchestrator owns the application state machine and coordinates all downstream services. It is a regular service, not the LLM. It triggers document processing, verification, fraud checks,

model scoring, property assessment, policy evaluation and decisioning in the right order, handles retries and timeouts, and enforces SLAs.

- Stage gates: fraud and KYC checks run before costly steps such as valuation and legal opinion.
- Parallelism: independent steps (bureau pull, bank statement analysis, legal and technical verification) run concurrently.
- Idempotency and resumability: every step is idempotent, and applicants can drop off and return without losing progress.
- Candidate implementations: a Python state machine on a durable queue, LangGraph, or Temporal if durable workflows are needed **[TBC]**.

5.3 Tool Gateway

The tool gateway is the security boundary between the LLM and the system. Every tool call is authenticated, authorised against the current session, validated against a schema, rate limited and logged.

- Session scoping is enforced here in code. The application ID comes from the authenticated session, never from LLM output.
- Tools are allow-listed per role. The applicant-facing agent has no tool that can approve, reject, change status or edit amounts.
- All inputs and outputs are written to the audit log with PII masked.

5.4 Document Intelligence

This pipeline converts uploaded files into validated, structured data with a confidence score.

1. **Ingest:** virus scan, format check, store the original immutably in the object store.
2. **Quality check:** resolution, skew, blur, completeness (missing pages). Ask the applicant to re-upload where needed.
3. **Classify:** identify document type (PAN, Aadhaar, salary slip, bank statement, ITR, GST return, sale deed, and so on).
4. **OCR:** dedicated OCR engine (for example PaddleOCR, docTR or Tesseract) **[TBC]**. Vision models may assist on layout-heavy documents after testing on real samples.
5. **Extract:** the LLM maps OCR text into a fixed JSON schema using Ollama's structured output at low temperature.
6. **Validate:** Pydantic schema checks plus deterministic rules (PAN format, date sanity, salary slip arithmetic, statement running-balance consistency).
7. **Cross-verify:** compare against authoritative sources (DigiLocker, CKYC, Account Aggregator, GST) and across documents.
8. **Route:** high-confidence, fully validated results proceed. Low-confidence or conflicting results go to a human verification queue.

Key documents and checks

Category	Documents	Representative checks
Identity / KYC	PAN, Aadhaar, address proof, photo	Format and checksum, name and date-of-birth match across documents, source verification
Income (salaried)	Salary slips, Form 16, bank statements	Slip totals and deductions add up, salary credits match statement, employer consistency

Category	Documents	Representative checks
Income (self-employed)	ITR, GST returns, financials, bank statements	Turnover reconciliation between GST, ITR and bank credits, seasonality, cash-flow patterns
Property	Sale deed, title chain, tax receipts, approved plan, occupancy / rental agreements	Owner name match, chain continuity, encumbrance status, plan approval, rental income evidence

5.5 Fraud and Anomaly Detection

Fraud detection runs early and again after key data arrives. It combines rules, statistical anomaly detection and supervised models when labelled fraud data exists.

Signal category	Examples	Technique
Identity	Synthetic or mismatched identity, PAN-Aadhaar mismatch, repeated use of contact details across applicants	Rules, entity resolution, graph links
Document forensics	Edited PDFs, inconsistent fonts or metadata, tampered figures, template mismatch	Metadata analysis, image forensics, template comparison
Income and bank behaviour	Inflated income, window-dressed balances, round-figure or circular credits, salary not matching slips	Rules plus anomaly models on transaction features
Property	Property already mortgaged (CERSAI), ownership mismatch, valuation far above comparables, forged title papers	External checks, valuation outlier detection
Velocity and network	Same property, phone, address, device or agent across multiple applications	Velocity rules, graph analytics

Output: a fraud score, a list of triggered signals with evidence, and a recommended action (clear, refer to fraud review, or reject). Confirmed cases are fed back as labels for future supervised models.

5.6 Credit Risk Models

Credit risk is expressed as Expected Loss, with each component modelled and reported separately:

$$EL = PD \times LGD \times EAD$$

Component	Meaning	Typical inputs
PD	Probability that the borrower defaults within a defined horizon (default definition, for example 90+ DPD, to be set by Risk) [TBC]	Bureau history and score, existing obligations, bank-statement conduct (bounces, average balance, inflow stability), income and employment stability, business vintage
LGD	Share of exposure lost if default occurs, after recovery from the property	Property type, location and liquidity, legal complexity, valuation quality and forced-sale discount, occupancy, expected time and cost to foreclose
EAD	Outstanding exposure at default	Sanctioned amount, tenure, repayment schedule (for term loans, close to outstanding principal)

Component	Meaning	Typical inputs
EL	Combined expected loss, used for risk-based pricing, LTV adjustment and referral	Output of the above

- **Model families:** gradient-boosted trees or scorecards for PD (explainable with reason codes). LGD may begin as a rule and grid-based approach and evolve to a regression model as recovery data accumulates.
- **Cold start:** if historical performance data is limited, start with bureau score plus rules, run models in **shadow mode**, and promote only after validation.
- **Explainability:** each score returns top contributing factors that map to approved reason codes.
- **Governance:** independent validation, calibration checks, documented limitations, and champion-challenger deployment.

5.7 Property Assessment

Property assessment feeds both the LTV calculation and the LGD model.

- **Legal verification:** title chain, encumbrances, CERSAI and registry checks, ownership and co-owner consent. Outcome recorded as clear, conditional or adverse with structured remarks.
- **Technical verification:** site inspection for existence, occupancy, construction status and adherence to approved plan.
- **Valuation:** market value and realisable (forced-sale) value from empanelled valuers. Policy may use the lower of two valuations [TBC].
- **Marketability:** a structured assessment of liquidity by property type and location, used as an LGD input.

Outputs are stored as structured fields (not free text) so that the policy engine and LGD model can consume them.

5.8 Policy Engine

The policy engine is a deterministic, configuration-driven service that computes eligibility. It takes verified income, obligations, property value, product parameters and risk outputs, and returns the eligible amount with a full calculation trace.

- All parameters (FOIR, LTV caps by property type, tenure limits, minimum and maximum ticket size, rate grids) are versioned configuration.
- Every run stores the policy version, inputs and step-by-step results so the outcome can be reproduced.
- Configuration changes require maker-checker approval and are audited.
- Implementation options: a Python rules module with strong test coverage, or a rules engine such as OPA or Drools [TBC].

The eligibility logic is described in Section 7.

5.9 Decision Service and Human Review

The decision service combines fraud outcome, property outcome, risk scores and eligibility into one of three outcomes, each with reason codes. Thresholds below are illustrative and must be calibrated by Credit Risk.

Condition	Outcome
Hard KYC failure, confirmed fraud, or adverse legal opinion	Reject (or escalate to fraud operations) with reason codes
Fraud score above refer threshold, or conflicting document signals	Refer to fraud / verification review
All checks pass, PD and EL within auto-approve band, amount within delegated authority	Auto-approve (conditional sanction subject to documentation)
EL in referral band, borderline FOIR / LTV, self-employed income needing judgement, or amount above delegated authority	Refer to underwriter
PD or EL above reject band, or eligible amount below product minimum	Reject with reason codes and, where possible, an alternative (lower amount, co-applicant, additional collateral)

The underwriter console shows the full picture: extracted data with source snippets, fraud signals, PD / LGD / EL breakdown, property findings, and the policy calculation trace. Overrides require a recorded reason and follow maker-checker rules.

5.10 Early Warning System (Post-Disbursement)

EWS monitors disbursed loans for early signs of stress. It is not part of the origination decision, and is planned as a later phase reusing the same data platform.

- **Signals:** EMI bounces and delayed payments, drop in account inflows (with consent), GST filing lapses, bureau deterioration, changes in property status or new encumbrances.
- **Output:** a stress score, watchlist tier and recommended actions (reminder, outreach, field visit, restructure review).
- **Additional use:** pre-approved and top-up offers to existing good-standing customers.

6. Application Lifecycle

6.1 End-to-End Flow

1. Applicant starts a chat. The agent captures profile, property and loan requirement.
2. Applicant uploads documents through secure upload links. Document intelligence extracts and validates data.
3. KYC verification and fraud pre-screen run. Failures stop or divert the case early.
4. Bureau pull and bank-statement / GST analysis produce income and behaviour features.
5. Property legal, technical and valuation workflows run in parallel.
6. PD, LGD, EAD and EL are computed. Fraud score is refreshed with the new data.
7. The policy engine computes FOIR-based and LTV-based limits and the final eligible amount.
8. The decision service returns auto-approve, refer or reject with reason codes.

9. Referred cases go to underwriters. Approved cases move to sanction and documentation.
10. The agent informs the applicant, using only tool-provided, approved wording.

6.2 Application States

State	Description	Next states
DRAFT	Conversation in progress, basic details captured	DOCS_PENDING, ABANDONED
DOCS_PENDING	Waiting for required documents	DOCS_PROCESSING
DOCS_PROCESSING	OCR, extraction and validation running	VERIFICATION, DOCS_PENDING
VERIFICATION	KYC, fraud screening and income verification	PROPERTY_ASSESSMENT, REJECTED, FRAUD_REVIEW
PROPERTY_ASSESSMENT	Legal, technical and valuation in progress	SCORING, REJECTED
SCORING	PD, LGD, EL and policy evaluation	DECISION
DECISION	Automated decision computed	APPROVED, UW_REVIEW, REJECTED
UW_REVIEW / FRAUD_REVIEW	Human review queues	APPROVED, REJECTED, VERIFICATION
APPROVED	Conditional sanction issued; hand-off to documentation and disbursement (outside this HLD)	HANDED_OFF, EXPIRED
REJECTED / ABANDONED / EXPIRED / HANDED_OFF	Terminal states	None

7. Eligibility Computation

The eligible amount is the minimum of the income-based limit, the collateral-based limit and the product cap, then adjusted for risk. All steps are deterministic.

7.1 Formulas

```

Max EMI           = (FOIR x Net Monthly Income) - Existing Obligations
Income-based      = Max EMI x [1 - (1 + r)^(-n)] / r
                  r = monthly interest rate, n = tenure in months
Collateral-based  = LTV cap x Applicable Property Value
                  (LTV cap by property type and risk band)
Eligible Amount   = min(Income-based, Collateral-based, Product cap)
                  then adjust for risk, round down to policy step

```

7.2 Worked Example (illustrative only)

Step	Value
Net monthly income	INR 1,50,000
FOIR limit / existing EMIs	55% / INR 20,000

Step	Value
Max new EMI	$(0.55 \times 1,50,000) - 20,000 = \text{INR } 62,500$
Rate and tenure	11% p.a., 15 years
Income-based limit	About INR 55 lakh
Property value and LTV cap	INR 1 crore at 60%
Collateral-based limit	INR 60 lakh
Eligible amount	$\min(55, 60) = \text{INR } 55 \text{ lakh}$ (FOIR is the binding constraint)

7.3 Design Considerations

- **Income recognition:** salaried income is straightforward. For self-employed applicants the method (cash profit, GST turnover, average bank balance, or a blend) must be configurable per policy [TBC].
- **Obligations:** taken from bureau and verified against bank statement debits, including undeclared EMIs.
- **Property value:** the applicable value follows policy, for example the lower of two valuations, or market versus realisable value.
- **Risk-based adjustment:** higher PD or EL can lower the LTV cap or FOIR, raise pricing, or trigger referral. The rule is configuration, not model output.
- **Trace:** the response includes each intermediate value so underwriters and applicants (through approved wording) can see what constrained the amount.

8. Data Architecture

8.1 Data Stores

Store	Contents	Candidate technology [TBC]
Operational database	Applications, applicants, properties, extracted fields, decisions, workflow state	PostgreSQL
Object store	Original documents and derived artefacts (immutable, versioned)	S3-compatible on-premises store such as MinIO
Feature store	Curated features for PD, LGD and fraud models, with point-in-time correctness	Feast or an internal feature layer
Model registry	Model versions, metrics, approvals, lineage	MLflow
Audit log	Append-only record of tool calls, decisions, overrides, configuration changes	Write-once store or table with hash chaining
Cache / queue	Sessions, task queues, rate limiting	Redis, plus a message queue as needed

8.2 Core Entities

Entity	Purpose
Application	The loan request, product, requested amount and tenure, current state

Entity	Purpose
Applicant / Co-applicant	Individual or entity profile, role, KYC status
Property	Address, type, owners, valuation records, legal and technical outcomes
Document	File reference, type, quality result, processing status
ExtractedField	Value, source document and location, confidence, verification status
FraudSignal	Rule or model that fired, severity, evidence
ModelScore	Model name and version, inputs snapshot, score, reason factors
PolicyRun	Policy version, inputs, calculation trace, eligible amount
Decision	Outcome, reason codes, decision maker (system or user), overrides
Conversation / AuditEvent	Masked transcripts and tool calls for review and compliance

8.3 Data Principles

- Every decision links to immutable snapshots of its inputs, so it can be replayed exactly.
- Point-in-time correctness for training data to avoid leakage.
- Retention and deletion schedules defined per data class in line with regulation and policy [TBC].
- Aadhaar, PAN and account numbers are masked in logs and non-essential displays.

9. Integration Architecture

External systems are accessed only through an integration layer with adapters, retries, circuit breakers, response caching where permitted, and full request/response audit. The list below is indicative and depends on vendor selection and commercial agreements.

System	Purpose	Status
Credit bureaus	Bureau report and score for PD features and obligations	[TBC] which bureaus
CKYC and DigiLocker	Identity and document verification	[TBC]
Account Aggregator framework	Consented bank statement and financial data	[TBC] AA / FIU set-up
GST and ITR sources	Turnover and income verification for self-employed applicants	[TBC]
CERSAI	Check for existing security interests on the property	[TBC]
Valuer and legal vendors	Property valuation, technical and legal reports	[TBC] empanelled vendors
Core banking / LMS	Sanction, disbursement and account creation after approval	[TBC]
Notification services	SMS, email and WhatsApp for status and consent flows	[TBC]

10. LLM Design and Guardrails (Ollama)

10.1 Deployment

- Ollama runs on-premises on GPU hardware [TBC]. The application calls it through its local API only.
- Pin exact model versions and quantisation. Upgrade through a controlled release with regression evaluation.
- Use separate model roles or instances for the chat agent and for document extraction so load and behaviour can be tuned independently.
- Set `OLLAMA\_NUM\_PARALLEL` and GPU memory limits to match expected concurrency. If concurrency demands grow, evaluate a higher-throughput serving option for production.
- Choose a model family that supports tool calling, and benchmark candidates (for example Llama, Qwen, Mistral families) on your own conversation and extraction tasks before committing.

10.2 Usage Patterns

Task	Approach	Controls
Conversation	Tool-calling agent with a fixed system prompt and allow-listed tools	Session-scoped tools, no decision authority, conversation length limits
Field extraction	OCR text into JSON schema via structured output, low temperature	Schema validation, deterministic cross-checks, confidence routing
Document classification	LLM or lightweight classifier with fixed label set	Unknown / low-confidence goes to human
Explanations	Narrate reason codes and tool results in plain language	Approved reason-code wording, no new numbers

10.3 Guardrails

- **Prompt injection:** document text and user messages are treated as data. Extraction prompts have no tool access. Only the chat agent has tools, and only the low-privilege ones listed in Section 5.1.
- **Numbers:** any figure in an LLM reply must match tool output. An output check flags amounts not present in tool results before the message is sent.
- **Scope control:** the agent declines topics outside the loan journey and does not give legal, tax or investment advice.
- **PII handling:** prompts contain only what the task needs. Logs store masked values.
- **Evaluation:** maintain a labelled, anonymised evaluation set of real documents and conversations. Re-run it on every model, prompt or schema change.

11. ML Lifecycle and Governance

Stage	Approach
Data and labels	Define default and fraud labels with Risk. Build point-in-time training sets. Document data lineage and known biases.

Stage	Approach
Development	Baseline scorecard or gradient-boosted model, with monotonic constraints where appropriate. Track experiments in the registry.
Validation	Independent validation of discrimination, calibration, stability and fairness before approval.
Deployment	Shadow mode first, then champion-challenger, then promotion. Versioned, rollback-ready.
Monitoring	Data drift, score distribution, approval and override rates, and realised performance versus prediction.
Retraining	Scheduled and trigger-based, with the same validation gates as initial release.

Cold start: If historical LAP performance data is limited, launch with a bureau-score-plus-rules approach and run ML models in shadow mode while data accumulates. Do not let unvalidated models drive auto-decisions.

12. Security, Privacy and Compliance

Regulatory mapping below is a starting point and must be confirmed with Compliance and Legal [TBC].

Area	Controls
Identity and access	SSO with role-based access control, least privilege, separate roles for applicant, underwriter, fraud ops, admin, and model ops. MFA for staff.
Data protection	Encryption in transit (TLS) and at rest. Keys in KMS or vault. Field-level encryption for Aadhaar, PAN and account numbers. Masking in UI and logs.
Data localisation	All storage and processing inside the organisation's infrastructure in India. External calls only to regulated data providers.
Privacy and consent	Capture and record purpose-specific consent, honour data-minimisation and retention rules under applicable law, including the Digital Personal Data Protection Act, 2023.
Digital lending norms	Align disclosures, consent, grievance redressal and data handling with applicable RBI digital lending directions.
Aadhaar handling	Follow UIDAI norms on collection, storage and display. Use masked numbers or reference tokens where permitted.
Model risk	Model inventory, validation, approval and monitoring consistent with the organisation's model risk policy.
Auditability	Append-only audit trail of decisions, overrides, configuration and tool calls. Reproducible decisions from stored snapshots.
Application security	Secure SDLC, dependency and container scanning, penetration testing, upload malware scanning, rate limiting and abuse detection.
Resilience	Backups, tested restore, disaster recovery objectives defined with the business.

13. Non-Functional Requirements

Targets are proposed starting points and must be agreed with the business [TBC].

Attribute	Proposed target
Availability	99.5% for the applicant chat and status services during business hours; graceful degradation to a form-based flow if the LLM is unavailable
Chat latency	First token within a few seconds under normal load; tool calls returning within their own SLAs
Document processing	Most standard documents processed within minutes of upload; long bank statements may take longer
Scalability	Horizontal scaling for stateless services. GPU capacity sized for peak concurrent chats plus extraction workload
Auditability	100% of decisions reproducible from stored inputs and versions
Accuracy	Extraction accuracy and fraud recall targets set per document type from the evaluation set
Maintainability	Policy and threshold changes through configuration without redeploy; modular services with clear APIs
Observability	Centralised logs, metrics, traces, and dashboards for model, workflow and LLM health

14. Deployment Architecture

- **Environments:** development, test / UAT with anonymised data, pre-production, and production, with separate credentials and data.
- **Runtime:** containerised services. Kubernetes or a comparable orchestrator is recommended for production [TBC]. Docker Compose is acceptable for a pilot.
- **GPU tier:** dedicated GPU node(s) for Ollama, isolated from CPU-only services. Consider separate nodes for chat and for batch extraction as load grows.
- **Network:** DMZ for the applicant-facing gateway, internal zones for services and data, restricted egress allowing only approved provider endpoints.
- **CI/CD:** automated tests for policy engine, extraction schemas and LLM evaluation set on every change. Signed images and controlled promotion.
- **Observability:** Prometheus and Grafana style metrics, centralised logging, and alerts on latency, error rates, drift and queue depth.

15. Proposed Technology Stack

Candidates only. Final choices depend on existing enterprise standards and skills [TBC].

Area	Candidate
LLM runtime	Ollama (existing), with a pinned tool-calling capable model
Backend services	Python with FastAPI
Workflow / agent orchestration	Custom state machine, LangGraph, or Temporal

Area	Candidate
OCR	PaddleOCR, docTR or Tesseract, with a vision model for hard layouts after testing
Validation	Pydantic plus rule modules
ML	scikit-learn, XGBoost / LightGBM, isolation-forest and graph-based methods for anomalies
Policy engine	Python rules module, or OPA / Drools
Data	PostgreSQL, MinIO, Redis, Feast (optional), MLflow
Identity and secrets	Keycloak or enterprise SSO, HashiCorp Vault or enterprise KMS
Observability	Prometheus, Grafana, centralised logging stack
Frontend	Web and mobile chat UI, React-based underwriter console

16. Key Risks and Mitigations

Risk	Impact	Mitigation
Poor OCR or extraction quality on bank statements and property papers	Wrong income and FOIR, wrong decisions	Dedicated OCR, validation rules, confidence routing to humans, continuous evaluation on real documents
Prompt injection through uploaded documents	Manipulated extraction or agent behaviour	Documents treated as data, no tools in extraction prompts, output validation, no decision authority for the LLM
LLM hallucinating amounts or policy	Compliance and customer-trust issues	Amounts only from tools, output number check, approved wording for explanations
Insufficient training data for PD / LGD	Weak or unstable models	Bureau-plus-rules baseline, shadow mode, external benchmarks, progressive rollout
Self-employed income misjudged	Over-lending or good-customer rejection	Configurable methods, conservative defaults, underwriter review for borderline cases
Valuation inflation or title fraud	Higher losses, weak LGD	Dual valuation, comparables outliers, CERSAI checks, legal opinion gates
GPU capacity and latency limits with Ollama	Slow chat, backlog in extraction	Capacity planning, separate workloads, queueing, evaluate higher-throughput serving if needed
Regulatory changes	Rework of flows and disclosures	Configuration-driven policy, regular compliance review, modular consent and disclosure components

17. Phased Roadmap

Indicative sequencing. Durations to be estimated after scoping.

Phase	Focus	Outcome
0. Foundations	Confirm product parameters, model and hardware, data access, integrations; set up platform, audit and access control	Agreed requirements and running platform
1. Pilot core	Chatbot onboarding, document upload and extraction for KYC and income, policy engine with FOIR and LTV, underwriter console	Assisted decisions with deterministic eligibility
2. Risk and fraud	Fraud rules and anomaly models, PD / LGD in shadow mode, property workflows and CERSAI checks	Risk-informed referral and auto-decision bands
3. Automation	Validated models drive auto-approve and reject within limits, risk-based LTV and pricing	Straight-through processing for clean cases
4. EWS and expansion	Post-disbursement monitoring, pre-approved and top-up offers, additional products	Portfolio-level monitoring

18. Open Decisions [TBC]

1. Which Ollama model(s) and what GPU hardware (type and VRAM) will be used, and whether chat and extraction share a node.
2. LAP product parameters: eligible property types, ticket size, tenure, rate grid, LTV caps by property type, FOIR limits.
3. Income assessment methods for self-employed applicants and how each is weighted.
4. Valuation policy: number of valuers, use of lower value, market versus realisable value.
5. Availability and quality of historical loan and performance data; default and fraud label definitions.
6. Choice of bureaus, Account Aggregator set-up, CKYC / DigiLocker access, CERSAI access, and valuer / legal vendors.
7. Delegated authority levels and the human review operating model (queues, SLAs, maker-checker).
8. Deployment target (on-premises Kubernetes or VMs), DR objectives and environment strategy.
9. Scope and timing of EWS, and whether pre-approved offers are in the first release.
10. Retention periods and consent architecture confirmed with Compliance and Legal.

Appendix A. Glossary

Term	Meaning
LOS	Loan Origination System
LAP	Loan Against Property
FOIR	Fixed Obligations to Income Ratio: share of income committed to fixed obligations
LTV	Loan to Value: loan amount as a percentage of property value
PD / LGD / EAD / EL	Probability of Default / Loss Given Default / Exposure at Default / Expected Loss
EWS	Early Warning System for detecting stress in existing loans
KYC / CKYC	Know Your Customer / Central KYC Registry
CERSAI	Central Registry of Securitisation Asset Reconstruction and Security Interest of India
AA	Account Aggregator framework for consented financial data sharing
OCR	Optical Character Recognition
DPD	Days Past Due
Maker-checker	Control where one person prepares a change and another approves it
Shadow mode	Running a model on live cases without using its output for decisions